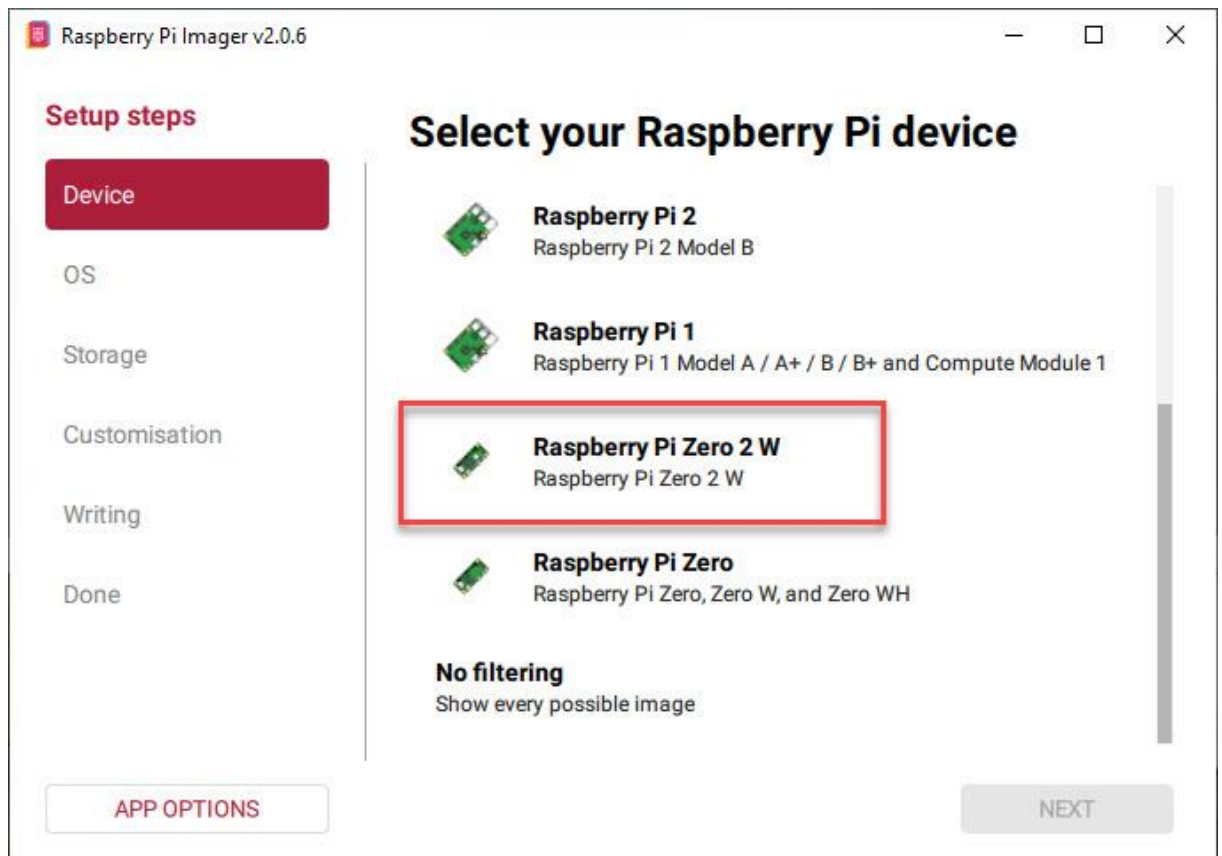


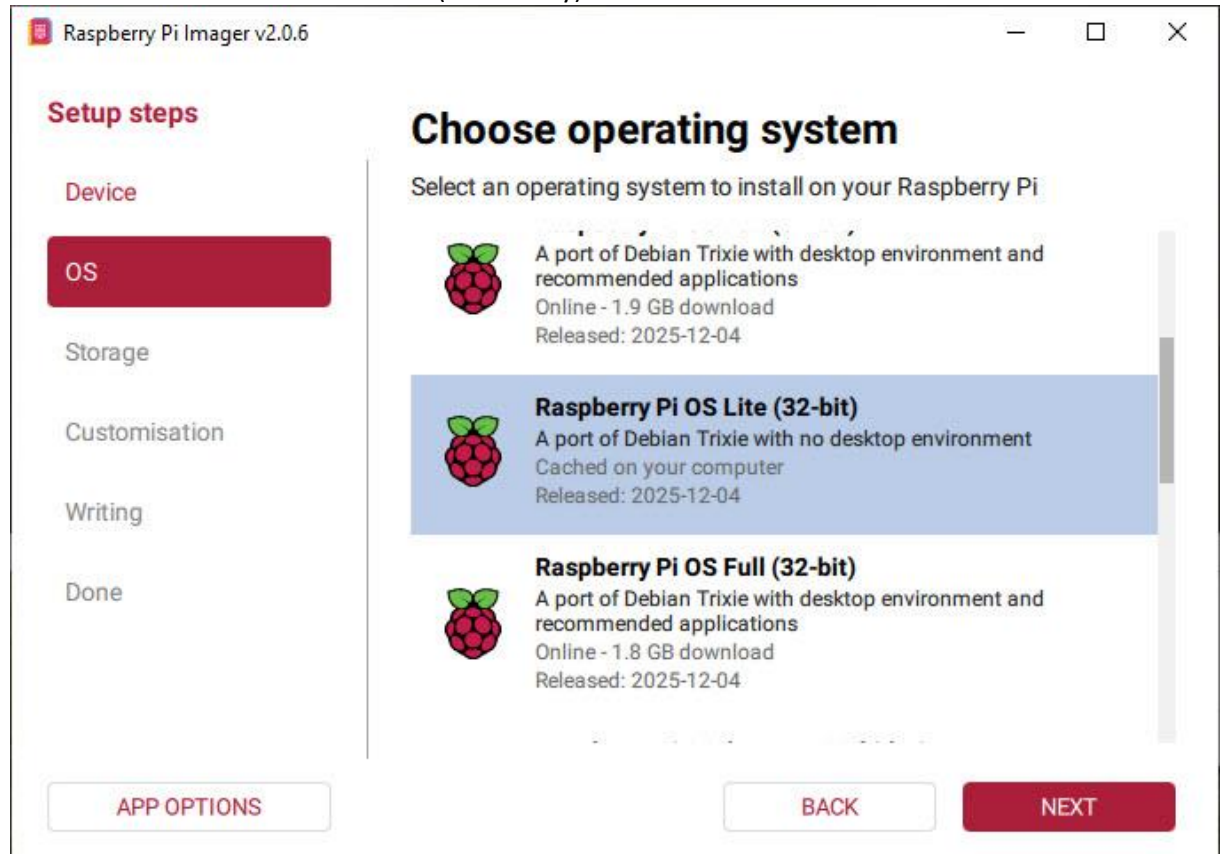
Pi-Ager install on Raspberry Pi 3B+/Pi 4/Pi 5/Pi zero (2)w under Pi OS 12 Lite Trixie 32Bit and 64Bit

- **For all Pi:**
Download and install Raspberry Pi Imager v2.0.6 or later from <https://www.raspberrypi.com/software/>.

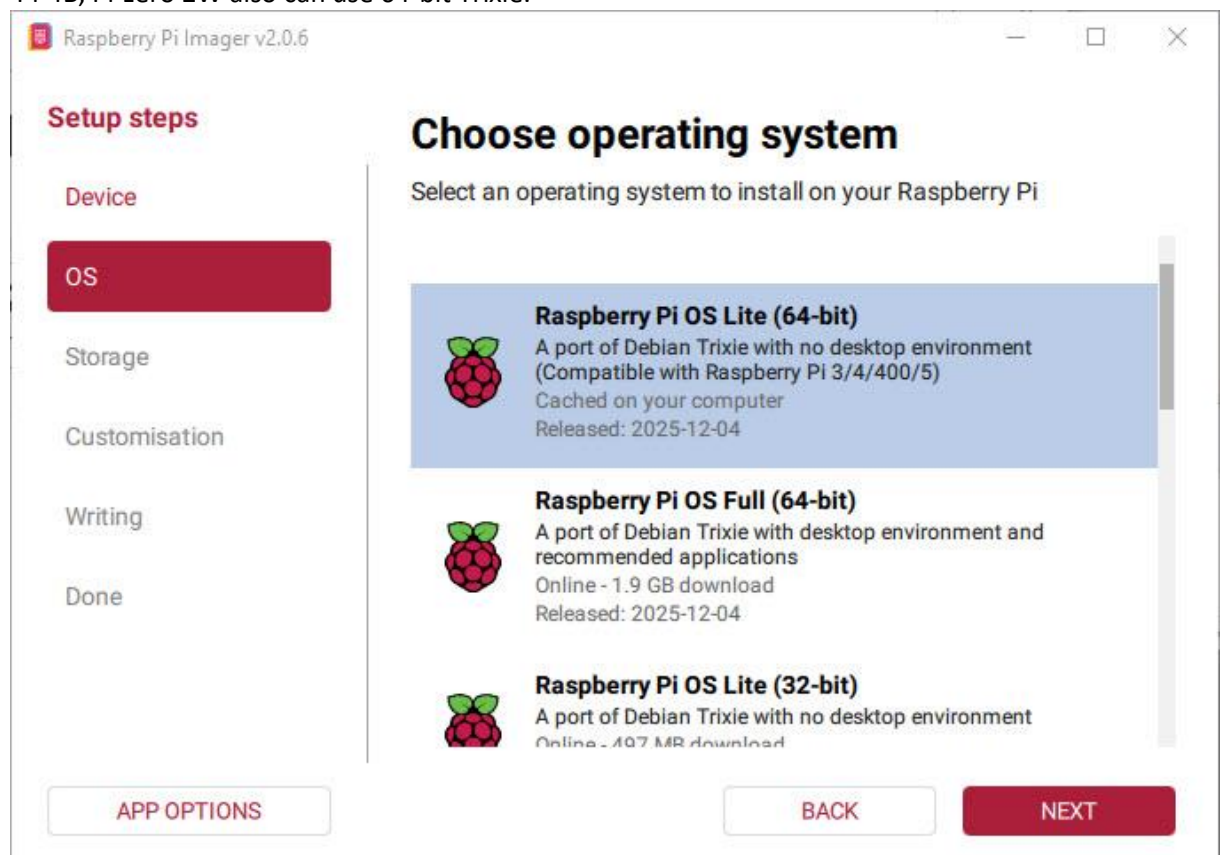
Start Raspberry Pi Imager, then continue with the follow steps:



Select Pi-OS 32 Bit Trixie for Pi Zero (32-bit only)



Or if you want to install pi-ager on Raspberry Pi-5 that needs a 64 Bit PI-OS, select Trixie. Pi-4B, Pi-zero 2W also can use 64-bit Trixie.



Raspberry Pi Imager v2.0.6

Setup steps

Device

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

APP OPTIONS

BACK

NEXT

Select your storage device



Generic STORAGE DEVICE USB Device
7.5 GB
Mounted as E:\

☒ Exclude system drives

Raspberry Pi Imager v2.0.6

Device

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

APP OPTIONS

SKIP CUSTOMISATION

BACK

NEXT

Customisation: Choose hostname

Enter your hostname

RaspberryPi

A hostname is a unique name that identifies your Raspberry Pi on the network. It should contain only letters, numbers, and hyphens.

Raspberry Pi Imager v2.0.6

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Customisation: Localisation

Select your location for suggested time zone and keyboard layout

Capital city: Berlin (Germany)

Time zone: Europe/Berlin

Keyboard layout: de

APP OPTIONS

SKIP CUSTOMISATION

BACK

NEXT

Raspberry Pi Imager v2.0.6

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

Customisation: Choose username

Create a user account for your Raspberry Pi

Username: pi

Password: Saved (hidden) — leave blank to keep

Confirm password: Re-enter to change password

The username must be lowercase and contain only letters, numbers, underscores, and hyphens.

APP OPTIONS

SKIP CUSTOMISATION

BACK

NEXT

Raspberry Pi Imager v2.0.6

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

Customisation: Choose Wi-Fi

SECURE NETWORK

OPEN NETWORK

SSID:

Network name

Password:

Network password
●●●●●●●●

Confirm password:

Re-enter password
●●●●●●●●

☐ Hidden SSID

APP OPTIONS

SKIP CUSTOMISATION

BACK

NEXT

Raspberry Pi Imager v2.0.6

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

Customisation: SSH authentication

Configure SSH access

Enable SSH

Learn about SSH

☒

Authentication mechanism:

☒ Use password authentication

☐ Use public key authentication

APP OPTIONS

SKIP CUSTOMISATION

BACK

NEXT

Raspberry Pi Imager v2.0.6

—□×

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

Customisation: Raspberry Pi Connect

Sign in to receive a token and enable Raspberry Pi Connect

Enable Raspberry Pi Connect

What is Raspberry Pi Connect?

☐

APP OPTIONS

SKIP CUSTOMISATION

BACK

NEXT

Raspberry Pi Imager v2.0.6

—□×

Device

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Write image

Review your choices and write the image to the storage device

Summary

Device:

Raspberry Pi Zero 2 W

Operating system:

Raspberry Pi OS Lite (32-bit)

Storage:

Generic STORAGE DEVICE USB Device

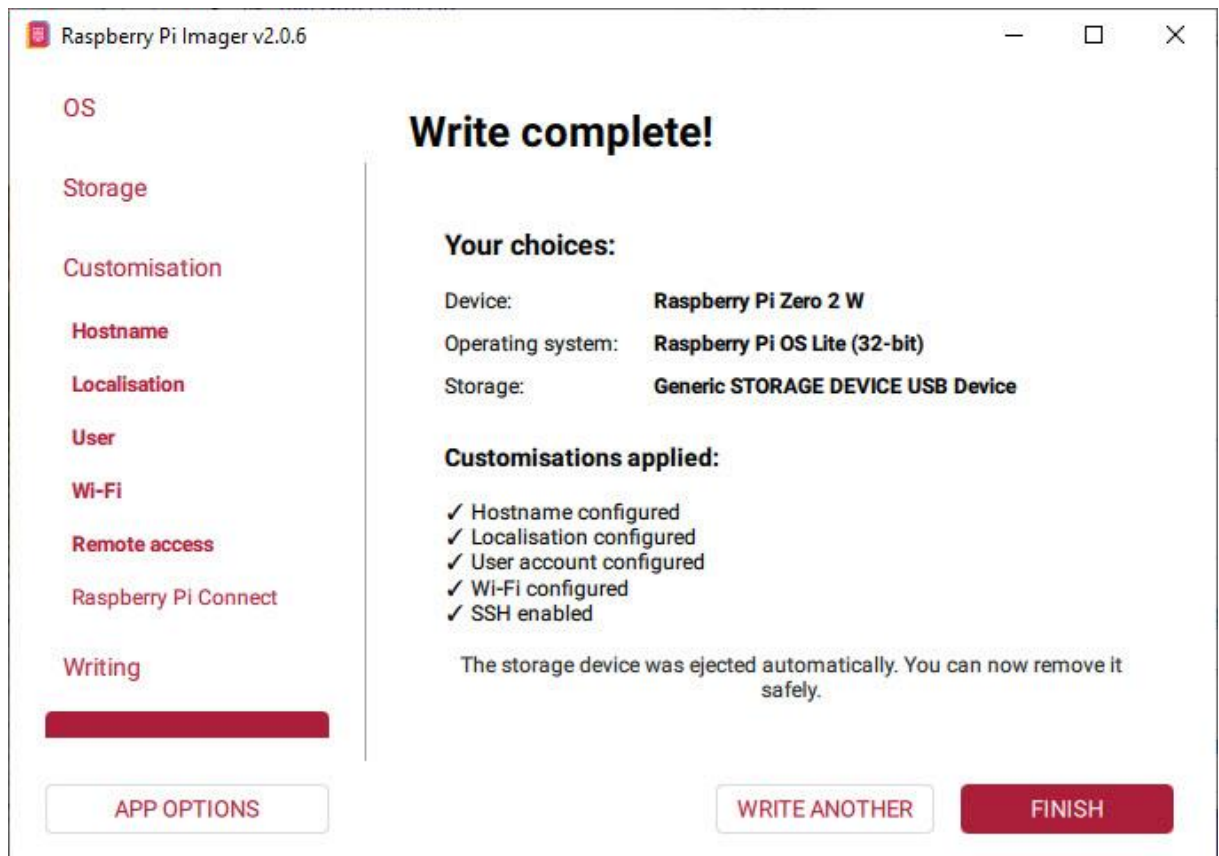
Customisations to apply:

- Hostname configured
- Localisation configured
- User account configured
- Wi-Fi configured
- SSH enabled

APP OPTIONS

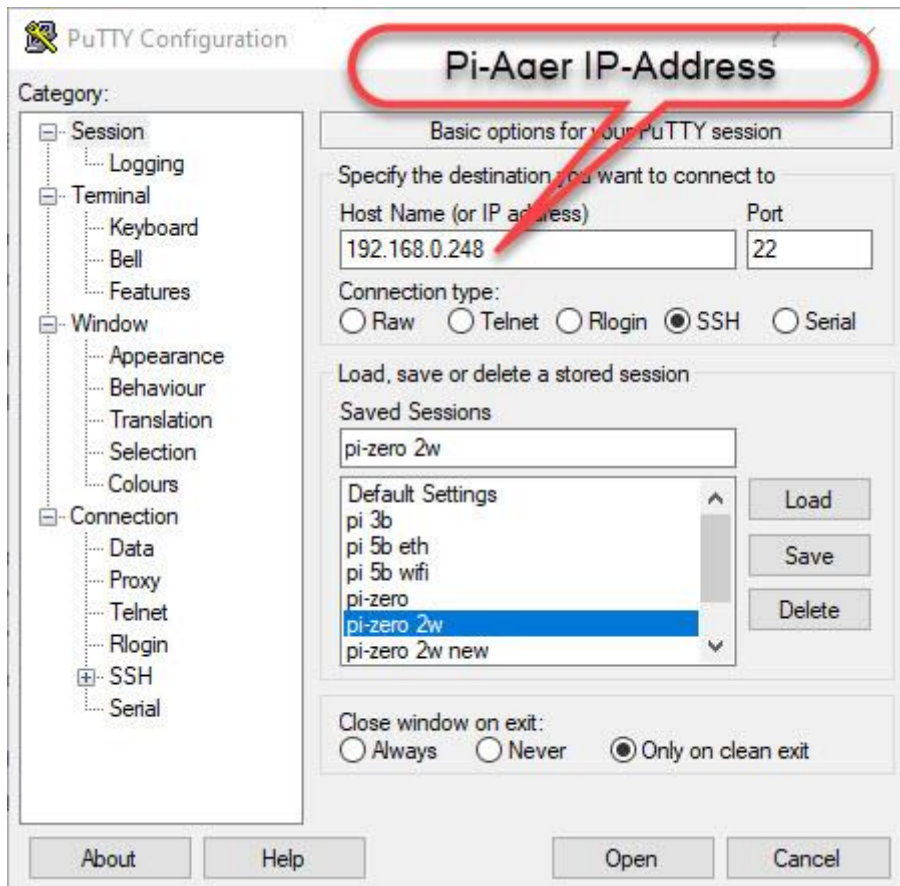
BACK

WRITE



Put your SD-Card in your Raspberry Pi and power on your Pi device.
Login via SSH (e.g. use PuTTY) or connect a HDMI monitor and USB keyboard to
your Raspberry device and continue with the setup as described below.

- Download and install PuTTY from <https://www.putty.org/>
- Optional download and install WinSCP from <https://winscp.net/eng/index.php>
- Start PuTTY to connect to Pi-Ager:



```

pi@pi-ager: ~
login as: pi
pi@192.168.0.206's password:
Linux pi-ager 6.6.74+rpt-rpi-v7 #1 SMP Raspbian 1:6.6.74-1+rpt1 (2025-01-27) arm
v7l

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Mar  5 20:00:41 2025 from 192.168.0.201

SSH is enabled and the default password for the 'pi' user has not been changed.
This is a security risk - please login as the 'pi' user and type 'passwd' to set
a new password.

pi@pi-ager:~ $

```

- Download system build scripts from github:
[wget -O pi-ager-build-system.sh -nv https://github.com/Tronje-the-Falconer/Pi-Ager/raw/refs/heads/resources/anleitungen/pi-ager-build-system.sh](https://github.com/Tronje-the-Falconer/Pi-Ager/raw/refs/heads/resources/anleitungen/pi-ager-build-system.sh)
[wget -O pi-ager-finalize-build.sh -nv https://github.com/Tronje-the-Falconer/Pi-Ager/raw/refs/heads/resources/anleitungen/pi-ager-finalize-build.sh](https://github.com/Tronje-the-Falconer/Pi-Ager/raw/refs/heads/resources/anleitungen/pi-ager-finalize-build.sh)

wget -O setup-wifi-ap.sh -nv <https://github.com/Tronje-the-Falconer/Pi-Ager/raw/refs/heads/resources/anleitungen/setup-wifi-ap.sh>

chmod +x *.sh

Start now first phase of Pi-Ager System setup:

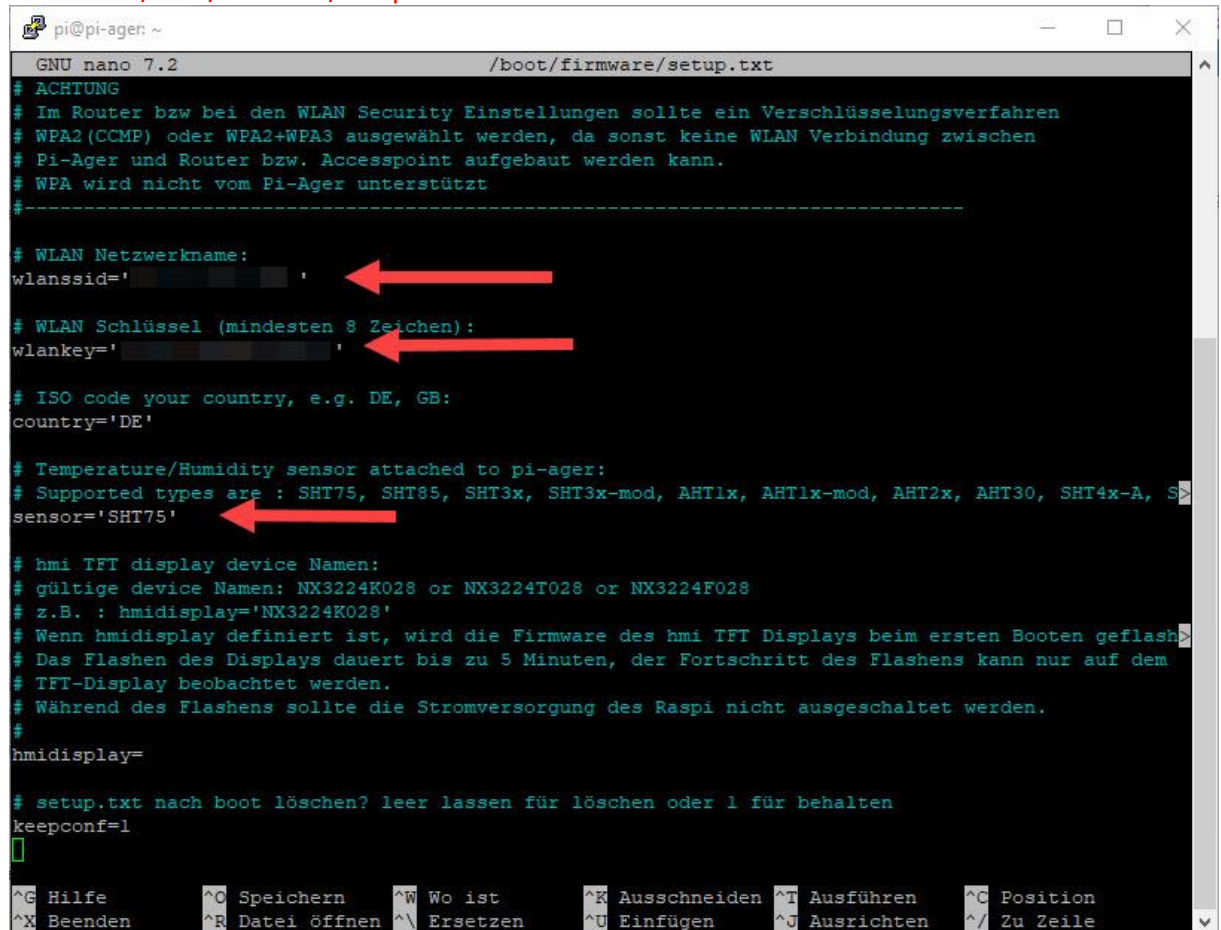
sudo ./pi-ager-build-system.sh

You can now follow the system setup, that is divided in 3 steps with rebooting between the steps. Restart the script after rebooting to continue the system setup :

sudo ./pi-ager-build-system.sh

After step 2 you are informed to edit the file /boot/firmware/setup.txt that contains information about passwords, WLAN setup parameter, temperature/humidity sensor to use and optional HMI display version connected to your Pi-Ager:

sudo nano /boot/firmware/setup.txt



```
GNU nano 7.2 /boot/firmware/setup.txt
# ACHTUNG
# Im Router bzw bei den WLAN Security Einstellungen sollte ein Verschlüsselungsverfahren
# WPA2 (CCMP) oder WPA2+WPA3 ausgewählt werden, da sonst keine WLAN Verbindung zwischen
# Pi-Ager und Router bzw. Accesspoint aufgebaut werden kann.
# WPA wird nicht vom Pi-Ager unterstützt
#-----
# WLAN Netzwerkname:
wlanssid=' '
# WLAN Schlüssel (mindesten 8 Zeichen):
wlankey=' '
# ISO code your country, e.g. DE, GB:
country='DE'
# Temperature/Humidity sensor attached to pi-ager:
# Supported types are : SHT75, SHT85, SHT3x, SHT3x-mod, AHT1x, AHT1x-mod, AHT2x, AHT30, SHT4x-A, S
sensor='SHT75'
# hmi TFT display device Namen:
# gültige device Namen: NX3224K028 or NX3224T028 or NX3224F028
# z.B. : hmidisplay='NX3224K028'
# Wenn hmidisplay definiert ist, wird die Firmware des hmi TFT Displays beim ersten Booten geflash
# Das Flashen des Displays dauert bis zu 5 Minuten, der Fortschritt des Flashens kann nur auf dem
# TFT-Display beobachtet werden.
# Während des Flashens sollte die Stromversorgung des Raspi nicht ausgeschaltet werden.
#
hmidisplay=
# setup.txt nach boot löschen? leer lassen für löschen oder 1 für behalten
keepconf=1
[
```

After saving setup.txt the last step initializes the Pi-Ager system with data from setup.txt by starting the script pi-ager-finalize-build.sh:

sudo ./pi-ager-finalize-build.sh

The system is then rebooted and ready to start the Pi-Ager. Open your browser window and enter the IP-Address of the Pi-Ager in the address line.

How to generate and deploy Pi-Ager Images

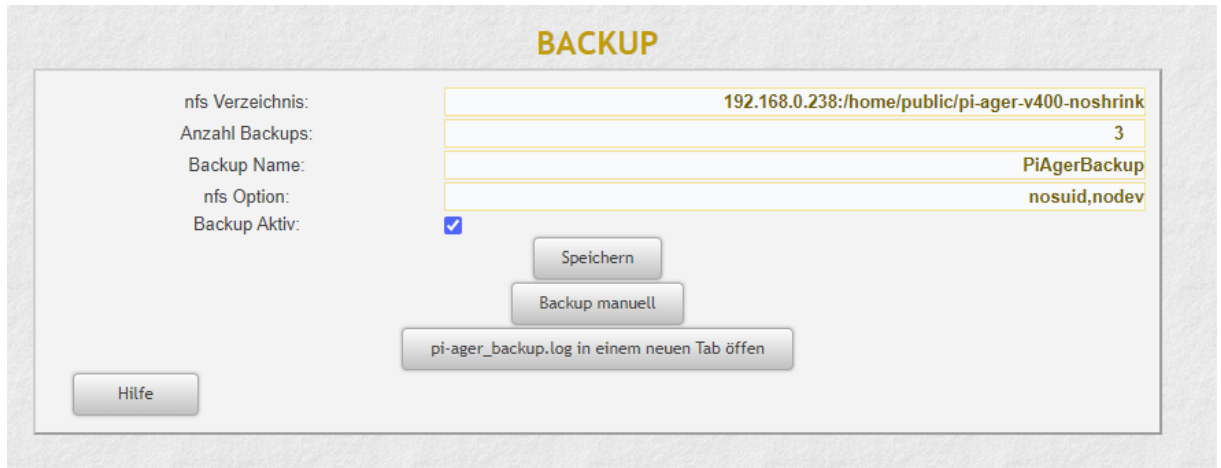
There are some steps to do for generating and deploying a new image from a running Pi-Ager system:

1. Setup a NFS Server to store a new image. The best way is using a Raspi 4B or 5 equipped with an USB Memory stick with 128GB or more. Using RPi 5 it is now possible to add a NVME SSD so that there will be enough space for storing one or more images.

How to setup a NFS Server :

<https://www.elektronik-kompendium.de/sites/raspberry-pi/2007061.htm>

2. On the Pi-Ager setup and start an Image backup on the ADMIN page. For example :



The screenshot shows the 'BACKUP' configuration page in the Pi-Ager ADMIN interface. It features a form with several input fields and a table. The 'nfs Verzeichnis:' field contains '192.168.0.238:/home/public/pi-ager-v400-noshrink'. The 'Anzahl Backups:' field contains '3'. The 'Backup Name:' field contains 'PiAgerBackup'. The 'nfs Option:' field contains 'nosuid,nodev'. The 'Backup Aktiv:' checkbox is checked. Below the form are three buttons: 'Speichern', 'Backup manuell', and 'pi-ager_backup.log in einem neuen Tab öffnen'. A 'Hilfe' button is located at the bottom left.

Field	Value
nfs Verzeichnis:	192.168.0.238:/home/public/pi-ager-v400-noshrink
Anzahl Backups:	3
Backup Name:	PiAgerBackup
nfs Option:	nosuid,nodev

3. Next start a terminal on the Pi-Ager and start the pi-ager_image.sh script with option -l: (lowercase 'L')

```
pi@pi-ager: /usr/local/bin

pi@pi-ager:~ $ cd /usr/local/bin/
pi@pi-ager:/usr/local/bin $ sudo pi-ager_image.sh -l
Backup ist inaktiv. Pi-Ager_image wird gestartet!
überprüfe ob der NFS-Server vorhanden ist.
Checking...
192.168.0.238:/home/public/pi-ager-v400-noshrink ist vorhanden
überprüfe ob Backup Pfad vorhanden ist.
Checking ...
/home/nfs/public ist vorhanden
überprüfe ob NFS Mount point definiert ist.
Checking ...
/home/nfs/public ist definiert
überprüfe ob NFS Mount point im file system angelegt ist.
Checking...
/home/nfs/public ist angelegt
NFSVOL=192.168.0.238:/home/public/pi-ager-v400-noshrink
NFSPATH=/home/nfs/public
NFSMOUNT=/home/nfs/public
umount: /home/nfs/public: not mounted.
hänge NFS-Volume 192.168.0.238:/home/public/pi-ager-v400-noshrink ein
mount with options: nosuid,nodev
mount NFS-Volume 192.168.0.238:/home/public/pi-ager-v400-noshrink successfull. Image script continues
.
Load last backup file.
Backup path is /home/nfs/public
Backup file is /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
Backup path with file is /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
Source File = /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
do_copy = false
my_image = false
Using /home/nfs/public/PiAgerBackup_2024-02-12-103529.img as source and target
#####
parted_output = 2:541065216B:15931539455B:15390474240B:ext4::;
partnum = 2
partstart = 541065216
loopback = /dev/loop0
#####
parted_output_boot = 1:4194304B:541065215B:536870912B:fat32::lba;
partnum_boot = 1
partstart_boot = 4194304
loopback_boot = /dev/loop1
#####
mount directory is /tmp/tmp.SfM3akt9CW
2024-02-12 10:39:10 URL:https://raw.githubusercontent.com/Tronje-the-Falconer/Pi-Ager/entwicklung/boo
t/firmware/setup.txt [3281/3281] -> "setup.txt" [1]
setup.txt copied to /tmp/tmp.SfM3akt9CW/boot/
cmdline.txt modified, added init=/usr/lib/raspberrypi-sys-mods/firstboot
rematch1 /tmp/
rematch2 tmp.SfM3akt9CW
change rootdir is tmp.SfM3akt9CW
/tmp
pi-ager_main.service disabled, will be enabled during setup_pi-ager.sh
Created symlink /etc/systemd/system/multi-user.target.wants/setup_pi-ager.service -> /lib/systemd/system/setup_p
i-ager.service.
setup_pi-ager service enabled
Running in chroot, ignoring command 'is-active'
hostname changed to pi-ager
unmount /tmp/tmp.SfM3akt9CW/boot
unmount /tmp/tmp.SfM3akt9CW
Detaching loop devices from /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
The image /home/nfs/public/PiAger_image_2024-02-12-103940.img was successfully created.
pi@pi-ager:/usr/local/bin $
```

4. Zip and deploy your new image.